

# Sabnaj Akter

Distributed Systems Engineer at AppsCode

✉ sabnajislam786@gmail.com 📞 +88 01777258450 🌐 GitHub 🌐 LinkedIn

## RESEARCH INTERESTS

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Distributed Systems, Cloud-Storage Reliability, Multi-Failure Resilience & Replication, Fault-Tolerant Systems, ML-for-Systems (Data-Driven Resource Management)

## EDUCATION

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**Khulna University of Engineering & Technology (KUET)**  
*BSc in Computer Science and Engineering (CGPA: 3.21/4.00)*

**Khulna, Bangladesh**  
*Jan 2020 – Apr 2025*

- **Relevant Coursework:** Operating Systems, Computer Networks, Database Systems, Machine Learning, Algorithms.

## WORK EXPERIENCE

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**Software Engineer**  
*AppsCode Inc. – Distributed Systems and Cloud-Native R&D*

**2025 – Present**  
*Dhaka, Bangladesh*

- Designed and developed cloud-native distributed database infrastructure systems for Kubernetes-based environments.
- Researched and developed Kubernetes database operators for IBM DB2 and DocumentDB.
- Worked on distributed orchestration systems for automated database lifecycle management.
- Built observability pipelines using Prometheus exporters, PromQL, and Grafana dashboards for distributed database infrastructure.
- Designed and implemented Raft-based failover and disaster recovery mechanisms to ensure low RPO/RTO guarantees and data durability in cloud-native database deployments.

## PROJECTS

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**KubeDB: IBM Db2 Operator** 🌐 🌐

*Go, Kubernetes, Cloud, TCP, Controller Runtime, Prometheus, Grafana*

- Implemented consensus-based failover and replication-aware recovery to ensure high availability and data durability under failure.
- Designed a TCP-based control plane for disaster recovery and cross-cluster communication.
- Automated E2E and chaos / fault-injection tests to validate reliability and recovery-latency guarantees under 1000+ database workloads and multi-failure scenarios.
- Researched and designed production-ready system architecture for product releases.
- Built monitoring exporters using PromQL metrics and Grafana dashboards for observability.
- Developed a Kubernetes operator for managing IBM Db2 database workloads in cloud-native environments.

**KubeDB: DocumentDB Operator** 🌐 🌐

*Go, Kubernetes, Raft, gRPC, Distributed Systems*

- Designed distributed database orchestration workflows for DocumentDB clusters on Kubernetes.
- Implemented Raft-based consensus and leader election mechanisms for high availability and fault tolerance.
- Designed gRPC-based internal services for cluster coordination and inter-node communication.
- Worked on automated provisioning, failover orchestration, and distributed database lifecycle management.

- Built reliability and recovery workflows for resilient distributed database operations.
- Measured failover recovery latency and durability-guarantee boundaries under large-scale fault injection and multi-failure scenarios.

## RESTful API Development

*Go, Docker, Backend Systems*

- Developed RESTful APIs using Go and Docker.
- Designed modular backend architecture for scalable services.

## PUBLICATIONS & RESEARCH

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### **BnLiteBait: Attention-Based Lightweight Clickbait Detection (Bengali)**

*Preprint (DOI: 10.13140/RG.2.2.31280.21766)*

*[ResearchGate Link](#)*

- Proposed a lightweight attention-based model for Bengali clickbait detection using headline-body fusion.
- Improved performance using ensemble bagging and t-SNE-based feature visualization analysis.

### **N-gram-Based Bengali Text Compression**

*Conference Paper — Under Review (ACM ICCA 2026)*

- Developed an N-gram-based Bengali text compression framework using statistical language modeling.
- Built tokenization and dictionary generation pipeline and evaluated compression performance.

## TECHNICAL SKILLS

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**Systems & Cloud Infrastructure:** Kubernetes, Docker, Linux, Kubernetes Operators, CRDs, Raft

**Networking, Backend & Observability:** TCP, HTTP, gRPC, REST APIs, TLS, mTLS, Prometheus, PromQL, Grafana

**Databases:** IBM Db2, PostgreSQL, DocumentDB

**Programming:** Go, Python, C/C++, Java, SQL, Bash

**Machine Learning & Data Methods:** Attention-Based Models, Statistical Language Modeling (N-gram), Ensemble Methods (Bagging), t-SNE Visualization

**Tools:** Git, GitHub, LaTeX

## ACHIEVEMENTS

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### **Competitive Programming**

*Codeforces Expert (1602), CodeChef 5-Star (2022)*

- Codeforces: Sabnaj423 (Expert, Max 1602)
- CodeChef: Sabnaj42 (5-Star, Max 2022)

### **Programming Contest Rankings**

*National and Regional Programming Contests*

- Intra KUET Programming Contest 2023: Ranked **2nd**
- National Girls' Programming Contest 2022: Ranked **8th**
- Ada Lovelace NGPC 2022: Ranked **10th**
- IUT IUPC 2024: Ranked **36th**

## LEADERSHIP & ACTIVITIES

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### **Vice President**

*Special Group Interested in Programming Contest (SGIPC), KUET*

**Feb 2024 – Feb 2025**